

MovieLOD

Linked movie data application · Semantic Web

From movie data to a knowledge graph

Ontology · RDF · Linked Open Data · SPARQL

Full 24-slide presentation

Team / members / class: [complete before submission]

30

real films

42

named classes

1,727

external links

Problem and five assignment requirements

Who did what in which film? Where did the data come from? How can we query it?

ID	Requirement	Deliverable / evidence
R1	Define an ontology	OWL/Turtle, classes, properties and restrictions
R2	Collect domain data	APIs, raw responses, URLs, timestamps and hashes
R3	Publish four-star data	RDF, HTTP IRIs, open license and publication
R4	Link to reach five stars	sameAs links, matching method and public data
R5	Provide SPARQL interfaces	Web application, endpoint and terminal

Required: report ≤15 pages · slides · 3–5-minute video. Running examples: Inception and Nolan.

Architecture and data flow

Asserted data, schema and inferred types are stored separately

01

COLLECT

Wikidata / DBpedia
collect.py → data/raw/

02

BUILD RDF

build.py → movies.ttl
+ ontology/movie.ttl

03

VALIDATE

validate.py / pytest
sources, fields and
queries

04

CLASSIFY

reason.py →
inferred_classes.ttl

APPLICATION: Web / Flask + RDFLib / Comunica / Terminal

Endpoint: movies.ttl. --reasoned: adds schema + inferred_classes.ttl.

Current dataset: interpreting the numbers

A selected sample of 30 films · evidence/statistics.json

30

films

851

people

1,010

contributions

45

companies

672

award entities

76/76

verified source hashes

19,339 data triples · 518 schema triples · 75 genres · 11 countries · 15 languages

R1 · Ontology: 42 classes in six groups

Direct reuse of dbo:Film, dbo:Person and dbo:Country

- 11 classes

CreativeWork / Film
- 8 classes

Agent / Person / Organization
- 6 classes

Contribution / ContributionRole
- 8 classes

Genre
- 5 classes

Award
- 4 classes

Country / Language / SourceSnapshot / Dataset

3 DBpedia classes + 39 ex: classes
14 classes receive types; 42 classes explicitly declared.

Ontology header:

Ontology IRI

https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology

Ontology Version IRI

e.g. https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology/1.0.0

Annotations +

rdfs:label

[language: en]

MovieLOD — movie ontology

dct:license

https://creativecommons.org/licenses/by-sa/4.0/

owl:versionInfo

2.0.0

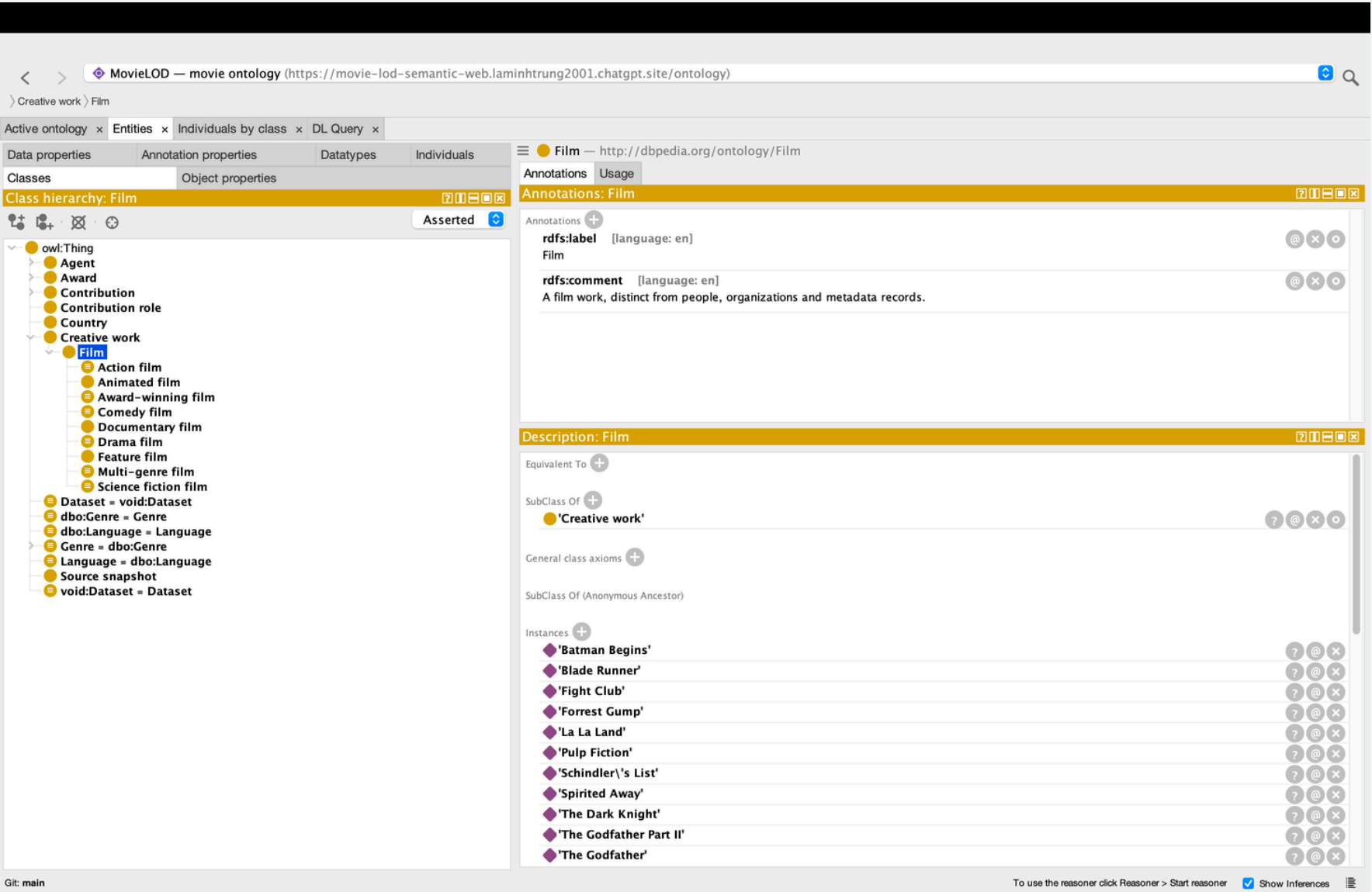
R1 · Creative works and film classes

Class definitions and additional instance types · (I) marks additional classification

CreativeWork

dbo:Film

- FeatureFilm · 29 films
- AnimatedFilm · 1 film
- DocumentaryFilm · 0 films
- ActionFilm (I)
- ComedyFilm (I)
- DramaFilm (I)
- ScienceFictionFilm (I)
- MultiGenreFilm (I)
- AwardWinningFilm (I)



(I): additional types. FeatureFilm/AnimatedFilm follow P31 mapping; DocumentaryFilm has no sample instances.

R1 · People, organizations and roles

A person may be both an Actor and a Filmmaker

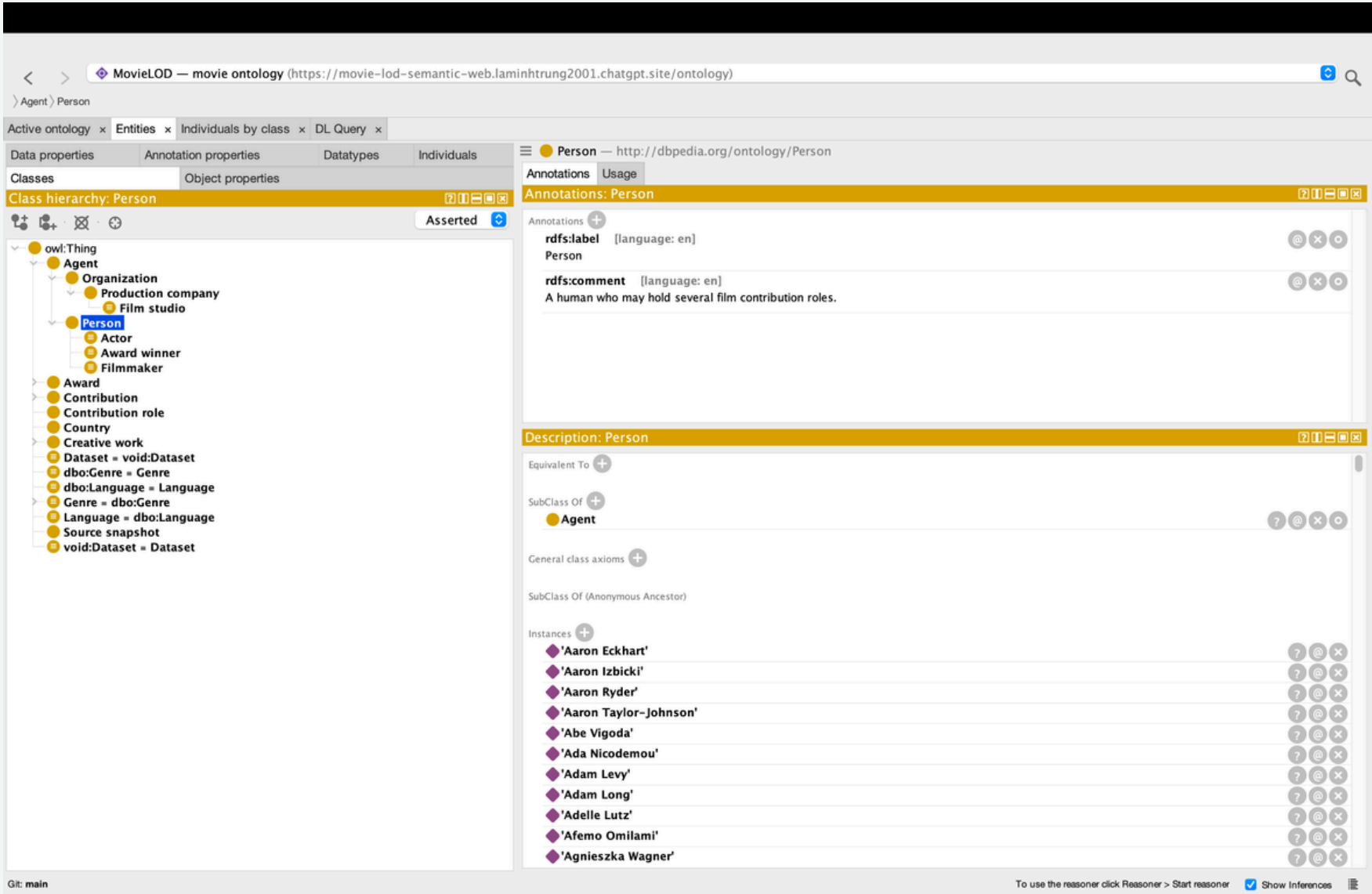
Agent

dbo:Person · 851 people

- Actor (I) · 769
- Filmmaker (I) · 89
- AwardWinner (I) · 290

Organization

- ProductionCompany · 45
- FilmStudio · 6 by counting; DL: 0



Person and Organization are different kinds of agent. Actor and Filmmaker may overlap.

R1 · Genres, awards and provenance

Each branch supports a specific group of questions

Genre

FictionGenre

- ActionGenre
- ComedyGenre
- DramaGenre
- ScienceFictionGenre

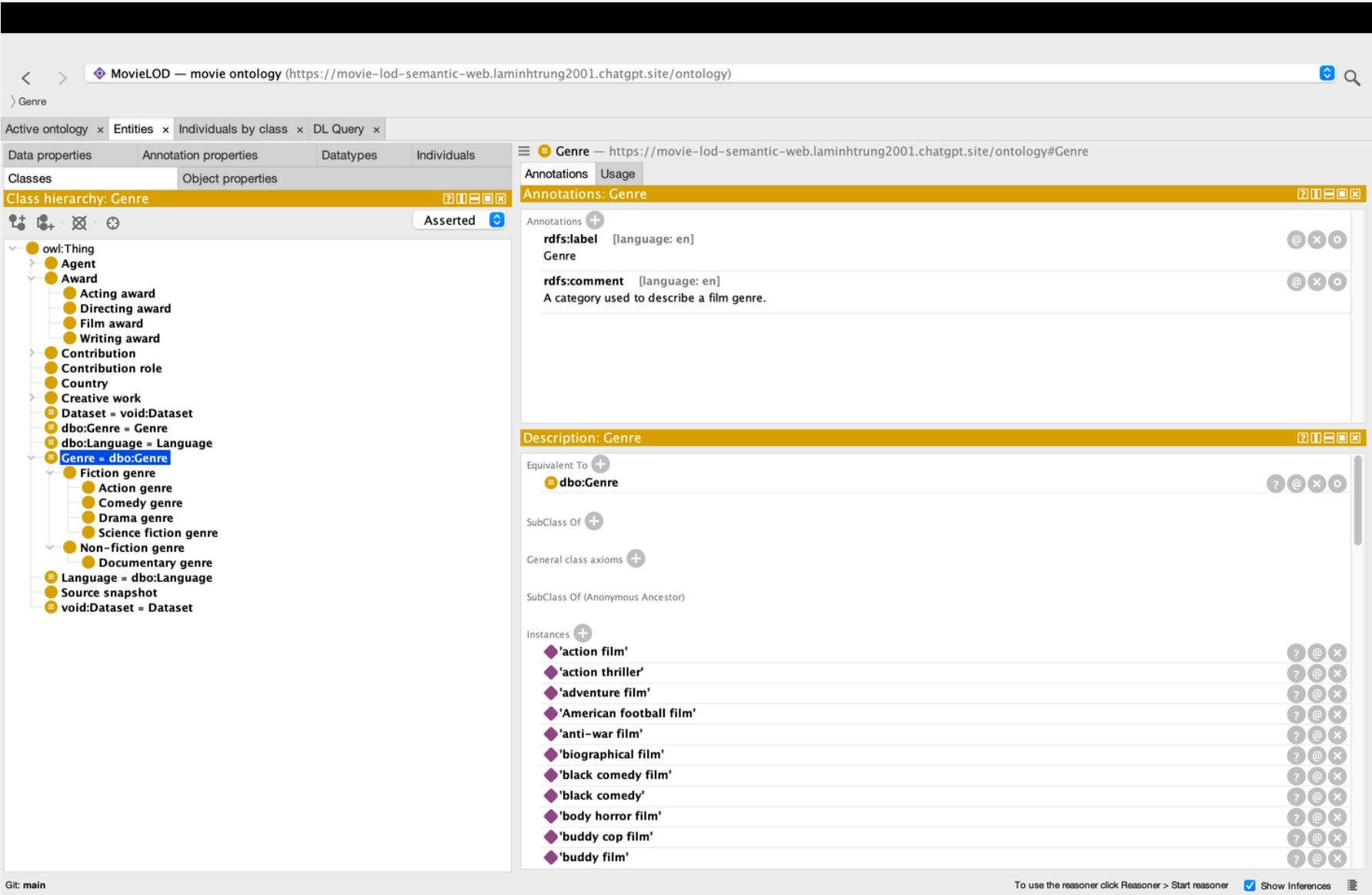
NonFictionGenre

- DocumentaryGenre

Award

FilmAward

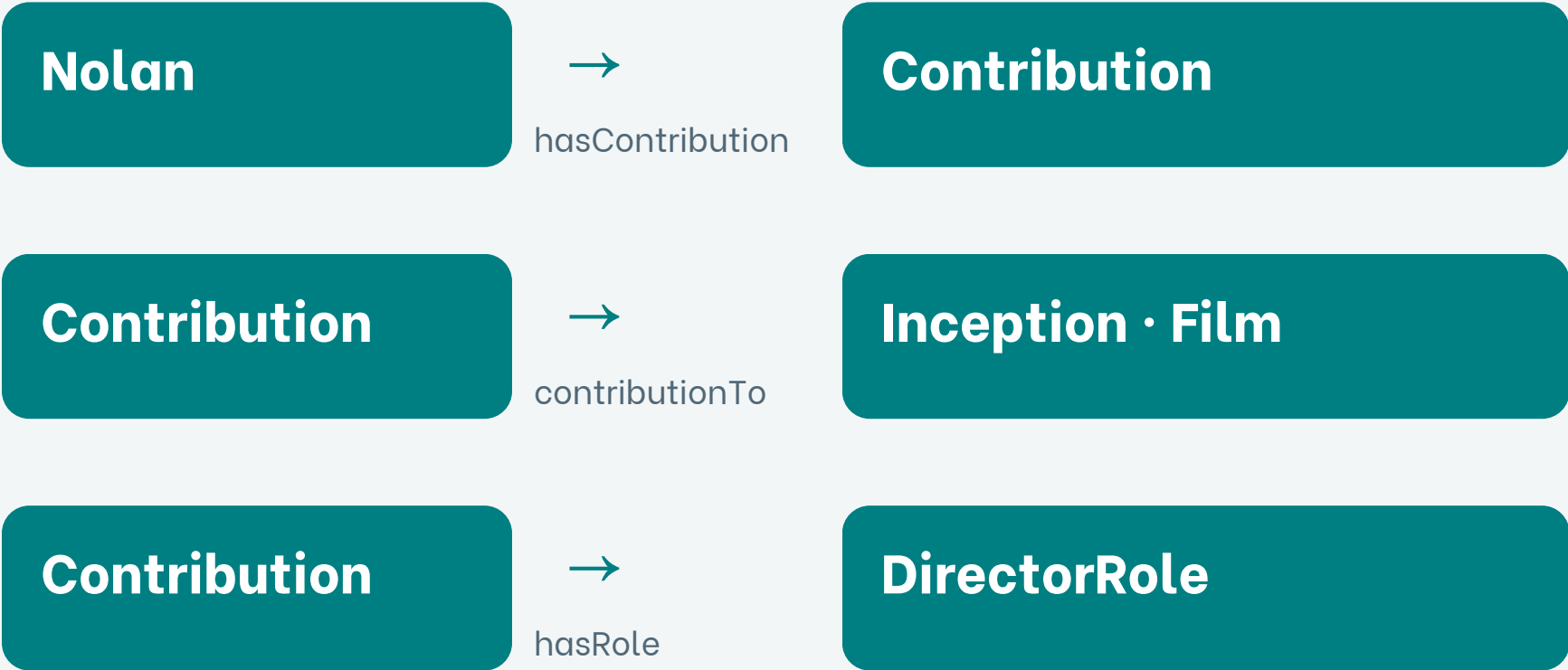
- ActingAward
- DirectingAward
- WritingAward



Supporting classes: dbo:Country · Language · SourceSnapshot · Dataset. Group mapping uses label keywords.

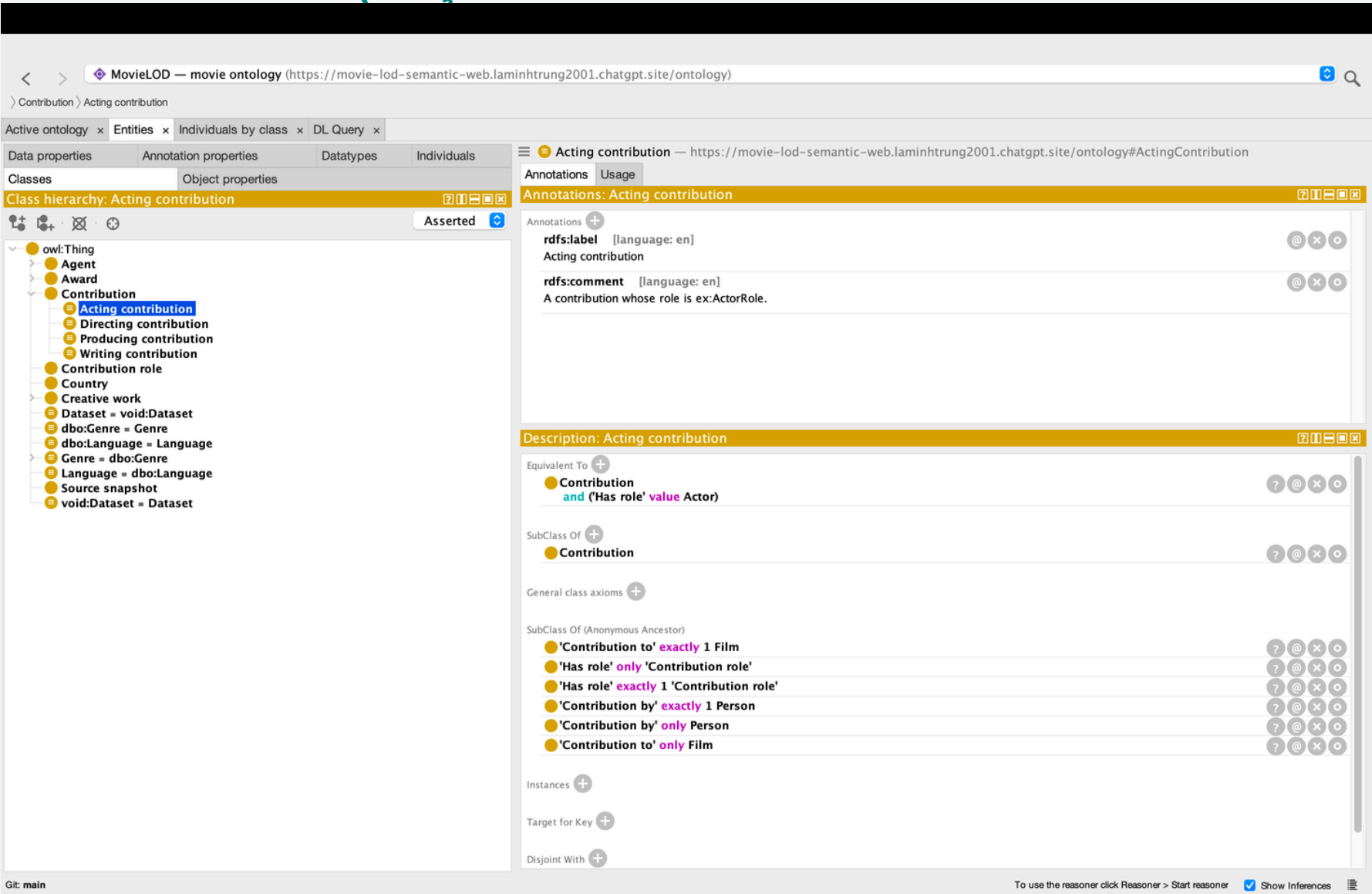
R1 · Contribution: who did what in a film?

One record connects exactly one person, one film and one role



Each role has its own record

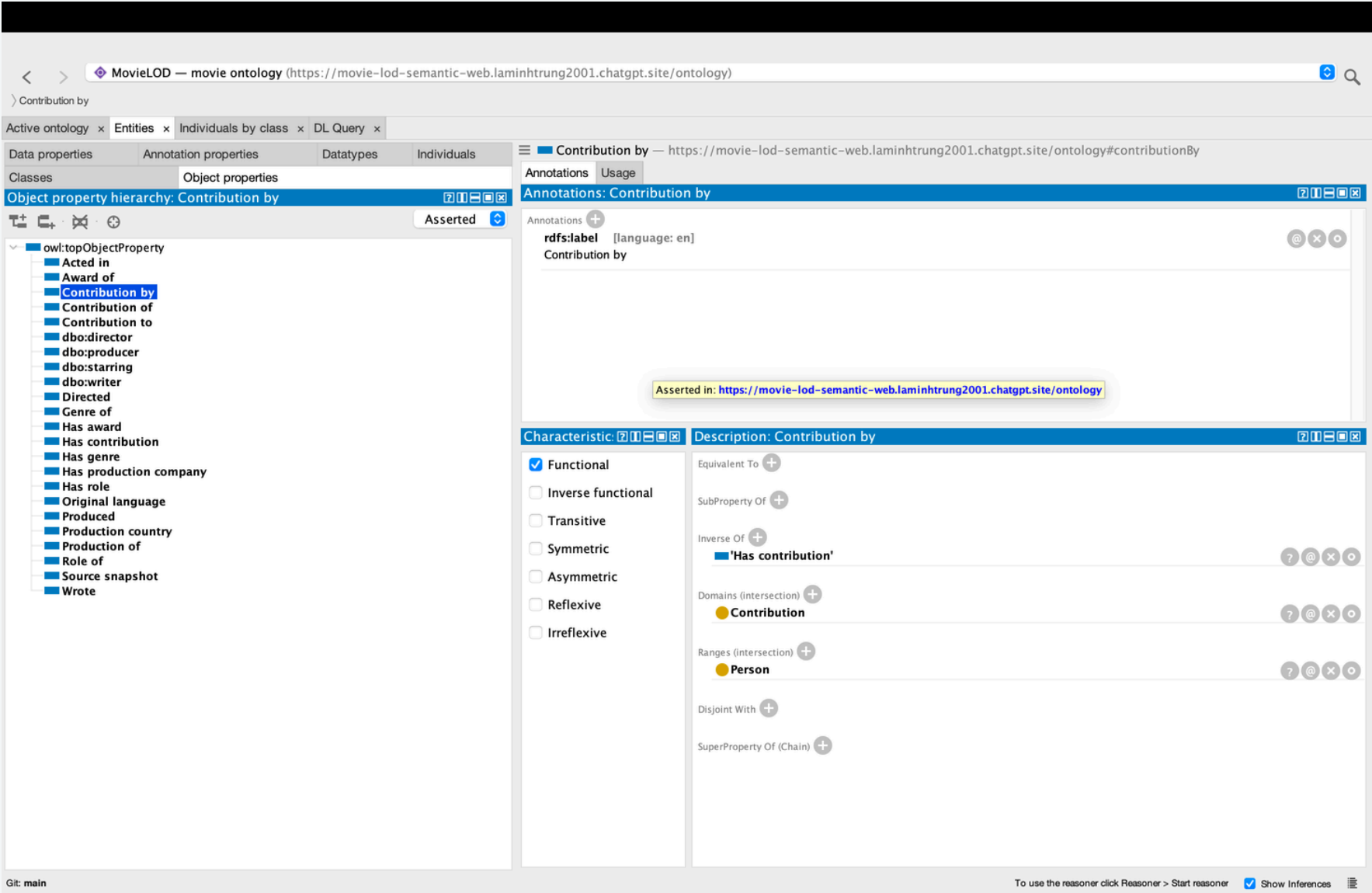
Director / Actor / Writer / Producer. Nolan has 3 roles in Inception.



R1 · Properties, domain/range and inverses

23 object properties · selected central relationships

Property	Domain → range	Inverse
contributionBy	Contribution → Person	hasContribution
contributionTo	Contribution → Film	contributionOf
hasRole	Contribution → Role	roleOf
dbo:director	Film → Person	directed
hasGenre	Film → Genre	genreOf
hasAward	Film or Person → Award	awardOf



contributionBy / contributionTo / hasRole: functional. Domain/range can infer types; inverse reverses a relation.

R1 • How OWL expresses meaning

Ontology constraints and application validation serve different purposes

Class definitions

equivalentClass + AND / OR / SOME
A Person with filmmaking contributions → Filmmaker.

Cardinality and functional

Contribution: exactly one person / film / role.
Functional: at most one value under OWL semantics.

Inverse and disjointness

director $\leftrightarrow$ directed; hasRole $\leftrightarrow$ roleOf.
Actor and Filmmaker are not disjoint.

Open-world assumption

Missing information $\neq$ negation.
Python validates required fields; OWL defines meaning.

R1 • Fourteen classes with additional types

None of these 14 types is asserted in the original data graph

Class	Condition in plain language	Count	Method
ActingContribution	Contribution with ActorRole	855	OWL RL
DirectingContribution	Contribution with DirectorRole	31	OWL RL
WritingContribution	Contribution with WriterRole	51	OWL RL
ProducingContribution	Contribution with ProducerRole	73	OWL RL
Actor	Person with acting contributions	769	OWL RL
Filmmaker	Person with filmmaking contributions	89	OWL RL
AwardWinner	Person who receives an award	290	OWL RL
ActionFilm	Film with ActionGenre	12	OWL RL
ComedyFilm	Film with ComedyGenre	4	OWL RL
DramaFilm	Film with DramaGenre	25	OWL RL
ScienceFictionFilm	Film with ScienceFictionGenre	6	OWL RL
AwardWinningFilm	Film that receives an award	26	OWL RL
MultiGenreFilm	Film with at least 2 Genre IRIs	30	IRI counting
FilmStudio	Company with at least 3 Film IRIs	6	IRI counting

12 classes: OWL RL, with matching Hermit/Pellet counts. Last 2: COUNT DISTINCT; DL instances: 0.

R1 · Inference example: Christopher Nolan

DirectorRole → DirectingContribution → Filmmaker

FACTS

Nolan a dbo:Person
c hasRole DirectorRole



RULE 1 · hasValue

c a DirectingContribution



RULE 2 · SOME + OR

Nolan a Filmmaker

The screenshot displays the MovieLOD ontology editor interface. The left pane shows the 'Class hierarchy: Filmmaker' with a tree structure including owl:Thing, Agent, Organization, Production company, Film studio, Person, Actor, Award winner, and Filmmaker. The right pane shows the 'Annotations: Filmmaker' section with rdfs:label 'Filmmaker' and rdfs:comment 'A person who directed, wrote or produced at least one film.' Below this, the 'Description: Filmmaker' section shows the equivalent class expression: `Person and (('Has contribution' some 'Directing contribution') or ('Has contribution' some 'Producing contribution') or ('Has contribution' some 'Writing contribution'))`. The bottom of the interface includes a status bar with 'Git: main' and a note to 'To use the reasoner click Reasoner > Start reasoner'.

Nolan: Person (asserted) → Person + Filmmaker + AwardWinner (with inferred types).

R1 · Cardinality: thresholds of two and three

Distinguish OWL definitions from counting distinct RDF terms

MultiGenreFilm

Film + min 2 hasGenre.Genre

Application: 30/30 films; DL: 0 inferred.

FilmStudio

ProductionCompany + min 3 productionOf.Film

Application: 6 companies; DL: 0 inferred.

Distinct IRIs ≠ provably different OWL individuals.

The screenshot displays the MovieLOD ontology editor interface. The left pane shows the 'Class hierarchy: Film studio' with a tree structure including 'owl:Thing', 'Agent', 'Organization', 'Production company', 'Person', 'Actor', 'Award winner', 'Filmmaker', 'Award', 'Contribution', 'Contribution role', 'Country', 'Creative work', 'Dataset', 'Language', and 'void:Dataset'. The right pane shows the 'Annotations: Film studio' section with two annotations: 'rdfs:label' (Film studio) and 'rdfs:comment' (A production company credited on at least three films in this dataset.). Below the annotations, the 'Description: Film studio' section shows the class hierarchy and the 'Equivalent To' section with the expression 'Production company and (Production of min 3 Film)'. The 'SubClass Of' section shows 'Production company' as a subclass of 'Production company'. The 'General class axioms' section is empty. The 'SubClass Of (Anonymous Ancestor)' section is empty. The 'Instances' section is empty. The 'Target for Key' section is empty. The 'Disjoint With' section is empty. The 'Disjoint Union Of' section is empty.

Local namespace filter + COUNT(DISTINCT IRI) avoids counting sameAs aliases as separate values.

R2 · Real collection and traceable sources

76/76 raw responses present · 76/76 SHA-256 hashes match

Resolve the right entity

Wikidata: exact Wikipedia sitelink.
DBpedia: matched root must be typed Film.

Collect relevant claims

P57 / P161 / P58 / P162: people and roles.
P166: awards; P272: companies; year/runtime/genre.

Trace and reproduce

URL + retrieval time + HTTP status + SHA-256.
Cached collect → build → validate rerun.

R2 · Actual Wikidata response and provenance

Read from the saved source file; SHA-256 verified. This is a source viewer, not a Protégé screenshot.

Response metadata

```
{
  "provider": "Wikidata",
  "url": "https://www.wikidata.org/w/api.php?action=wbgetentities&format=json&languages=en&sites=enwiki&titles=Inception%7CInterstellar+%28film%29%7CThe+Dark+Knight%7CMemento+%28film%29%7CDunkirk+%282017+film%29%7CThe+Prestige%7CBatman+Begins%7CTenet%7CThe+Godfather%7CThe+Godfather+Part+II&props=labels%7Cdescriptions%7Csitelinks%7Cclaims",
  "retrieved_at": "2026-10-06T08:43:21.455429+00:00",
  "sha256": "56d8b57c83cb1ad3dc2d2e0cf47035801fed896cc0dca296a6e8ff1e0c246e1"
,
  "path": "data/raw/1db5b6a369cc85140c7afe5e.json",
  "http_status": 200
}
```

Selected raw fields (references omitted)

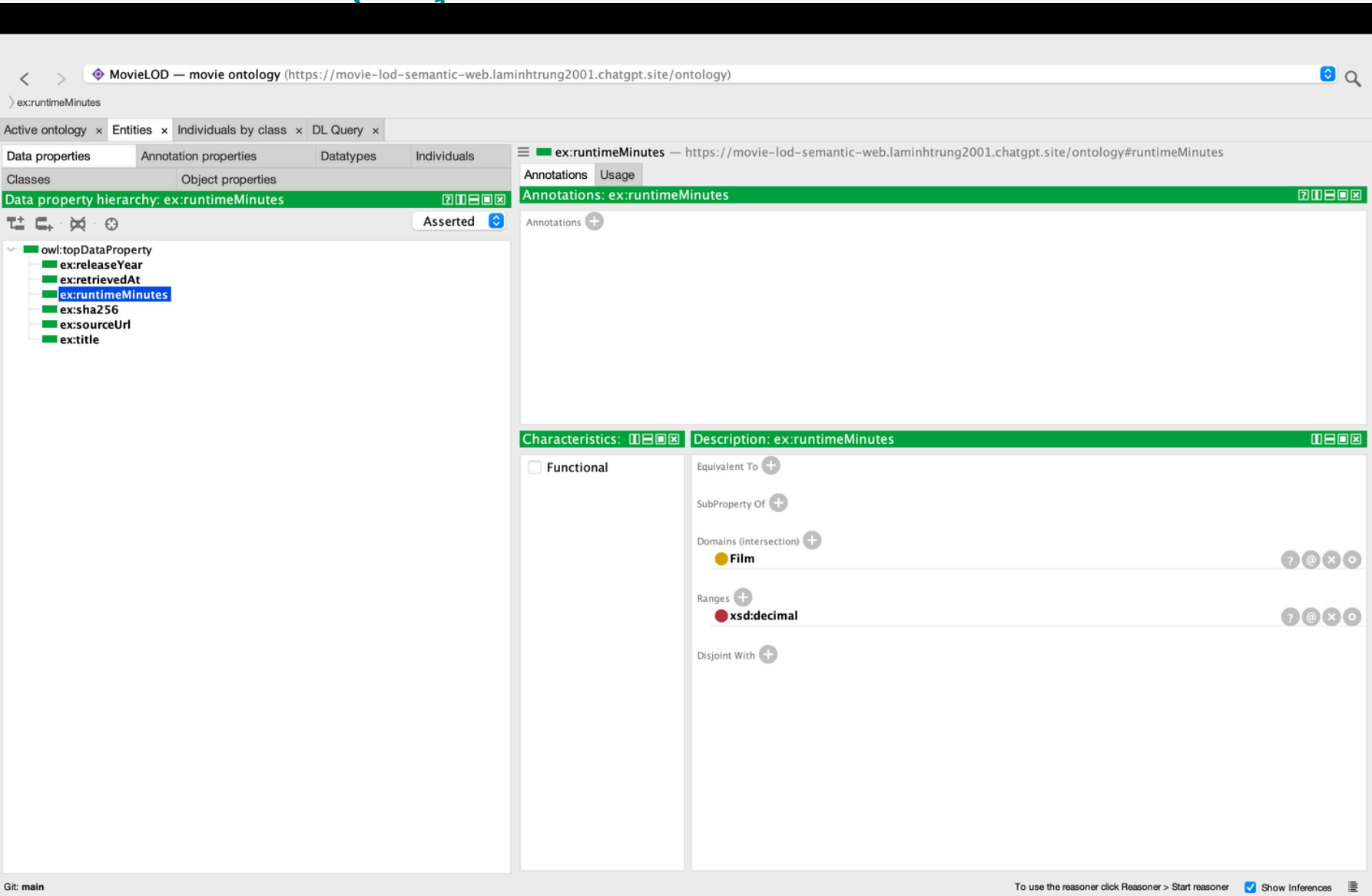
```
{
  "id": "Q25188",
  "labels": {
    "en": {
      "language": "en",
      "value": "Inception"
    }
  },
  "P57_director": [
    {
      "mainsnak": {
        "snaktype": "value",
        "property": "P57",
        "hash": "c163ceb80831ed3f753fb3b90ec8c31d153eb0bf",
        "datavalue": {
          "value": {
            "entity-type": "item",
            "numeric-id": 25191,
            "id": "Q25191"
          },
          "type": "wikibase-entityid"
        },
        "datatype": "wikibase-item"
      }
    }
  ]
}
```

Right: actual Wikidata metadata and response excerpt. Hashes verify integrity, not real-world truth.

R3 · RDF and typed literal values

19,339 data triples · 518 schema triples · Turtle / JSON-LD / OWL

```
res:film-Q25188 a dbo:Film ;
ex:title "Inception" ;
ex:releaseYear 2010 ;
ex:runtimeMinutes "148.0"^^xsd:decimal ;
dbo:director res:person-Q25191 .
```



Data property	xsd: type
title	string
releaseYear	integer
runtimeMinutes	decimal
sourceUrl	anyURI
retrievedAt	dateTime
sha256	string

R3–R4 · From RDF to Linked Open Data

Star levels are cumulative · publication and links are verified

1 star	Public data + open license	CC BY–SA 4.0
2 stars	Structured data	Films, people, roles and attributes
3 stars	Open formats	CSV / Turtle / JSON–LD
4 stars	HTTP IRIs + RDF + lookup	HTML, JSON–LD and RDF descriptions
5 stars	Links to external datasets	1,699 Wikidata + 28 DBpedia

sameAs: shared identity. sourceSnapshot: provenance. All four checked public RDF graphs match local data.

R3 · Looking up Inception’s IRI

One identifier offers readable and machine-readable descriptions

← MovieLOD

Inception

<https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/resource/film-Q25188>

Download RDF (Turtle)

Property	Value
director	Christopher Nolan
producer	Emma Thomas
producer	Christopher Nolan
starring	Dileep Rao
starring	Michael Caine
starring	Elliot Page
starring	Joseph Gordon-Levitt
starring	Tohoru Masamune
starring	Claire Geare
starring	Cillian Murphy
starring	Tom Hardy
starring	Ken Watanabe
starring	Pete Postlethwaite
starring	Taylor Geare
starring	Talulah Riley
starring	Tom Berenger

MovieLOD — movie ontology (https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology)

Creative work > Film > Feature film

Active ontology x Entities x Individuals by class x DL Query x

Data properties

Annotation properties

Datatypes

Individuals

Classes

Object properties

Individuals: Inception

Inception

Inception — actor — Cillian Murphy

Inception — actor — Claire Geare

Inception — actor — Dileep Rao

Inception — actor — Earl Cameron

Inception — actor — Elliot Page

Inception — actor — Joseph Gordon-Levitt

Inception — actor — Ken Watanabe

Inception — actor — Leonardo DiCaprio

Inception — actor — Lukas Haas

Inception — actor — Marion Cotillard

Inception — actor — Michael Caine

Inception — actor — Michael Gaston

Inception — actor — Pete Postlethwaite

Inception — actor — Silvie Laguna

Inception — actor — Talulah Riley

Inception — actor — Taylor Geare

Inception — actor — Tohoru Masamune

Inception — actor — Tom Berenger

Inception — actor — Tom Hardy

Inception — actor — Virgile Bramly

Inception — actor — Yuji Okumoto

Inception — director — Christopher Nolan

Inception — producer — Christopher Nolan

Inception — producer — Emma Thomas

Inception — writer — Christopher Nolan

independent film

Independent Spirit Award for Best Director

Independent Spirit Award for Best Female Lead

Independent Spirit Award for Best First Feature

Independent Spirit Award for Best Male Lead

Independent Spirit Award for Best Screenplay

Independent Spirit Award for Best Supporting Female

Independent Spirit Award for Best Supporting Male

Independent Spirit Robert Altman Award

India Catalina Awards

Indiana Film Journalists Association Award for Best Director

Annotations

Usage

Annotations: Inception

Annotations

rdfs:label [language: en]

Inception

prov:wasDerivedFrom

<https://www.wikidata.org/wiki/Q25188>

Description: Inception

Types

'Feature film'

Film

Same Individual As

dbpedia:Inception

Q25188

Different Individuals

Property assertions: Inception

'Has production company' 'Legendary Pictures'

'Has production company' 'Syncopy Inc.'

'Has production company' 'Warner Bros. Entertainment'

'Has production company' 'Warner Bros. Pictures'

'Original language' English

'Original language' French

'Original language' Japanese

'Production country' 'United Kingdom'

'Production country' 'United States'

'Source snapshot' 'DBpedia source snapshot'

'Source snapshot' 'Wikidata source snapshot'

Data property assertions

ex:releaseYear 2010

ex:runtimeMinutes 148.0

ex:title "Inception"

Negative object property assertions

Negative data property assertions

Git: main

To use the reasoner click Reasoner > Start reasoner ☒ Show Inferences

Left: real application screenshot. Right: pending Protégé evidence for Inception in the main Knowledge Graph.

R5 · Querying Inception on the Web

SELECT → title, year, runtimeMinutes and director

```
SELECT ?title ?year ?runtimeMinutes ?director
WHERE {
  ?film a dbo:Film ; ex:title "Inception" ;
  ex:title ?title ; dbo:director ?person .
  ?person rdfs:label ?director .
  OPTIONAL { ?film ex:releaseYear ?year }
  OPTIONAL { ?film ex:runtimeMinutes ?runtimeMinutes }
}
```

2010 · 148 minutes · Nolan

SELECT chooses columns; WHERE matches; OPTIONAL preserves rows.

Use the PREFIX declarations in queries/02_inception.rq. Actual result: 2010 · 148 minutes · Christopher Nolan.

MovieLOD

QueriesFilmsOntologyDataset

Download RDF ↗

SEMANTIC WEB / MOVIES

Ask questions about movie data.

301,72742

filmsexternal linksclasses

Explore films with source records, stable identifiers and links to Wikidata and DBpedia.

SPARQLQuery language

Sample queries

Inception: year, runtime and director

PREFIX ex: <https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?title ?year ?runtimeMinutes ?director WHERE {
 ?film a dbo:Film ; ex:title "Inception" ; ex:title ?title ;
 dbo:director ?person .
 ?person rdfs:label ?director .
 OPTIONAL { ?film ex:releaseYear ?year }
 OPTIONAL { ?film ex:runtimeMinutes ?runtimeMinutes }
}

Run queryCtrl ⌘

Copy

SELECT: get a table · ASK: test a condition · CONSTRUCT: create an RDF graph.

Results

Download results

1 result · 0.01 s

title	year	runtimeMinutes	director
Inception	2010	148.0	Christopher Nolan

Querying the application SPARQL endpoint.

Reading an RDF statement: Inception → director → Christopher Nolan. ex: is the project vocabulary; dbo: is the DBpedia vocabulary; variables start with ?.

COLLECTED DATA

Films in the dataset

Find a film

Film title or director

Data checked: 08 Oct 2026 · Sources: project code and evidence/

19 / 24

R5 · One person, multiple roles and questions

The screenshot filters Nolan: Director, Writer and Producer in Inception

MovieLOD

QueriesFilmsOntologyDataset

Download RDF ↗

SEMANTIC WEB / MOVIES

Ask questions about movie data.

Explore films with source records, stable identifiers and links to Wikidata and DBpedia.

301,72742

filmsexternal linksclasses

SPARQL

Query language

Sample queries

Inception: year, runtime and director

PREFIX ex: <https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?personName ?roleName WHERE {
 ?film ex:title "Inception" ; ex:contributionOf ?contribution .
 ?contribution ex:contributionBy ?person ; ex:hasRole ?role .
 ?person rdfs:label ?personName . ?role rdfs:label ?roleName .
 FILTER(STR(?personName) = "Christopher Nolan")
} ORDER BY ?roleName ?personName

Run query ⌘ ↵Copy

SELECT: get a table · ASK: test a condition · CONSTRUCT: create an RDF graph.

Results

Download results

3 results · 0.03 s

personName	roleName
Christopher Nolan	Director
Christopher Nolan	Producer
Christopher Nolan	Writer

Querying the application SPARQL endpoint.

Reading an RDF statement:

Inception → director → Christopher Nolan. ex: is the project vocabulary; dbo: is the DBpedia vocabulary; variables start with ?.

COLLECTED DATA

Find a film

Film title or director

25 records

All Inception contributions; the image shows only Nolan.

8 / 4 / 7

Nolan films / Inception companies / Godfather awards.

14 Web samples / 24 files

Hierarchy requires schema; inferred types require --reasoned.

Data checked: 08 Oct 2026 · Sources: project code and evidence/

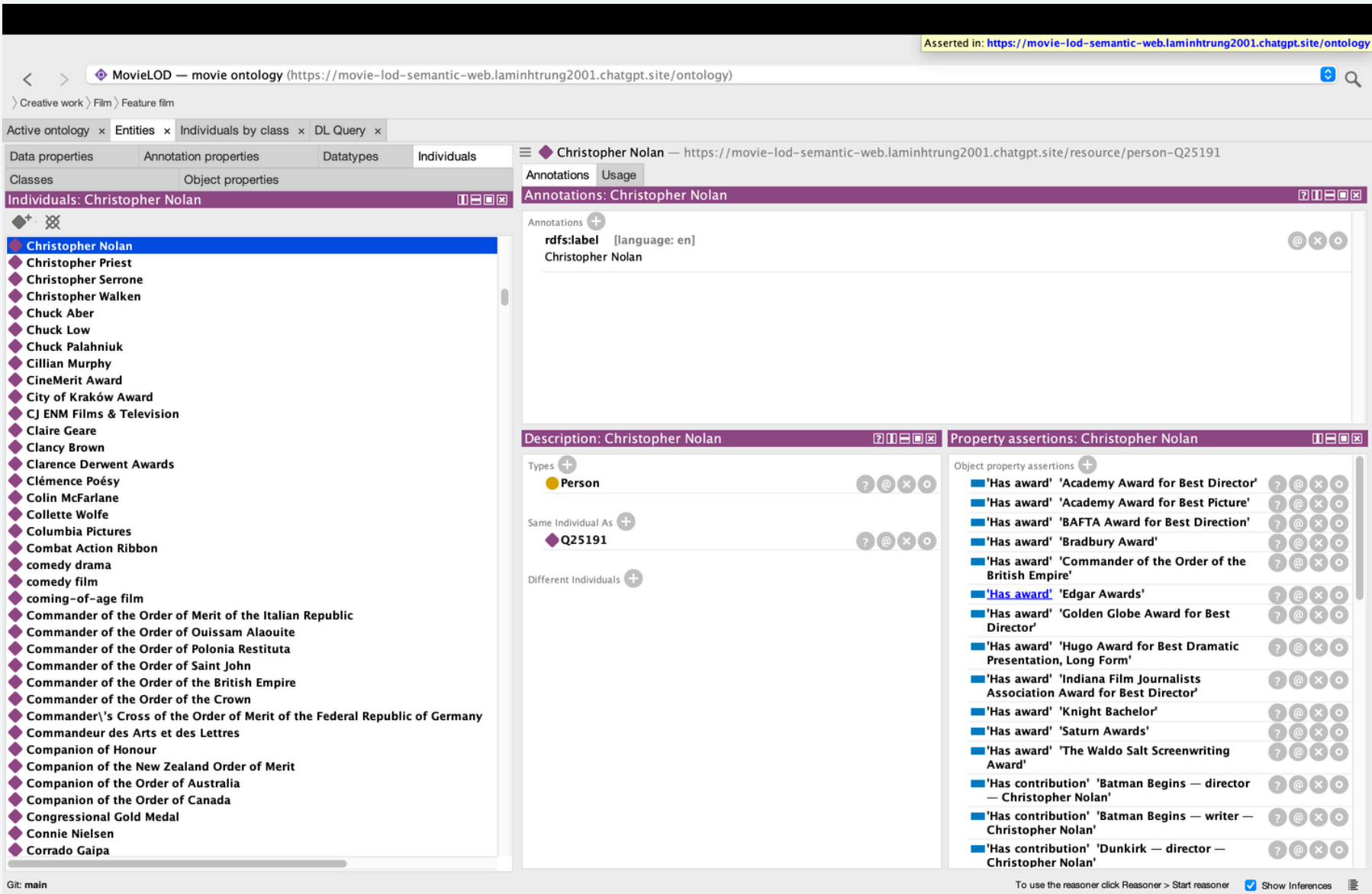
20 / 24

R5 · Endpoint, terminal and inferred types

Same Nolan query: one asserted type → three with saved inference

```
curl -X POST http://127.0.0.1:8000/sparql \
-H "Content-Type: application/sparql-query" \
--data-binary @queries/02_inception.rq

.venv/bin/python src/query.py \
queries/24_nolan_asserted_types_only.rq
.venv/bin/python src/query.py \
queries/24_nolan_asserted_types_only.rq --reasoned
```



Mode	Nolan types returned
Asserted graph	dbo:Person
With inference	Person + Filmmaker + AwardWinner

--reasoned loads schema + saved types. Endpoint: asserted graph by default. SELECT/ASK: JSON; graph queries: Turtle.

Validating data, application and publication

Execution evidence is stored in evidence/

76/76

Matching source hashes

24

Query files executed

14 + 9

Tests + browser checks

8 / 4

Public HTTP 200 / graph match

```
MovieLOD · Actual verification results

14 passed, 2 warnings in 96.97s (0:01:36)

Hermit 1.3.8.1099 · Movie_Knowledge_Graph.owl
Consistent: Yes · unsatisfiable named classes: 0

{
  "ActingContribution": 855,
  "DirectingContribution": 31,
  "WritingContribution": 51,
  "ProducingContribution": 73,
  "Actor": 769,
  "Filmmaker": 89,
  "AwardWinner": 290,
  "ActionFilm": 12,
  "ComedyFilm": 4,
  "DramaFilm": 25,
  "ScienceFictionFilm": 6,
  "AwardWinningFilm": 26,
  "MultiGenreFilm": 0,
  "FilmStudio": 0
}

Data checks: True
Source hashes: True · 76/76
SPARQL files executed: 24

Sources: evidence/tests.txt, hermit_run.json, validation.json
This is an execution-log viewer, not the Protégé interface.
```

14 tests / 9 browser checks. Hermit/Pellet verify DL separately; the image shows actual execution logs.

Assessment and limitations

Evidence for 5/5 requirements · 10/10 is a proposed self-score

Req.	Evidence	Score
R1	Ontology + OWL + tests	2/2
R2	76 sources + hashes + reproducibility	2/2
R3	Public RDF matches local data	2/2
R4	1,727 published identity links	2/2
R5	Web / endpoint / terminal	2/2

Total 10/10 · proposed self-score, not an official grade

Data limitations

Selected 30-film sample; label-based groups.
Hashes ≠ truth; budget/streaming data absent.

Reasoning limitations

12 OWL RL classes; 2 IRI-counting rules.
HermiT/Pellet: consistent; threshold classes: 0.

Semantic Web value

Shared IRIs, vocabulary, links and provenance.
SQL can also answer many of these questions.

Conclusion and pending Protégé evidence

24 slides · five-page report · four-minute-fifty-second demo

Image	Slide	Evidence to capture
P00	5	Ontology IRI and version
P01	6	Film class hierarchy
P02	7	People and organizations
P03	8	Genre and award hierarchies
P04	9	Contribution restrictions
P05	10	Domain, range, inverse and functional
P06	13	Filmmaker definition
P07	14	FilmStudio cardinality
P08	16	Runtime datatype
P09	18	Inception individual
P10	21	Nolan asserted types

Screenshot slots are placeholders, not captured evidence. Full English narration is in Speaker Notes.

Before submission

1. Complete team / member details.
2. Insert 11 genuine screenshots.
3. Save them in evidence/protege/ for automatic replacement.
4. Rehearse with the speaker script and review the video.